

Nirbhay Singh

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Summary

AI/ML Engineer with hands-on experience building Retrieval-Augmented Generation (RAG) pipelines, LLM-powered applications, transformer-based NLP systems, recommendation engines, and scalable backend APIs. Experienced in Machine Learning, Deep Learning, Natural Language Processing, Semantic Search, Vector Databases, and Generative AI solutions using LangChain, FastAPI, FAISS, ChromaDB, HuggingFace Transformers, TensorFlow, and AWS. Skilled in prompt engineering, vector embeddings, model deployment, and end-to-end AI application development. Seeking opportunities as a **Software Engineer, Machine Learning Engineer, AI Engineer, Generative AI Engineer, or LLM Engineer**.

Experience

TATA BlueScope Steel Pvt. Ltd

AI/ML Intern

Jamshedpur

May 2025 - Jul 2025

- Developed a FinBERT-based NLP pipeline processing 500+ financial news articles daily at approximately 82% sentiment accuracy.
- Built Streamlit dashboards visualizing sentiment trends across 20+ stocks, reducing manual analysis effort by 60%.
- Implemented automated ingestion, preprocessing, transformer inference, and reporting workflows.

Anatron International Pvt. Ltd

Software Engineer

Jamshedpur

Feb 2026 - Present

- Develop and maintain production REST APIs using Node.js, Express.js, FastAPI, and React.js.
- Improved frontend performance by 30% through lazy loading and optimized API communication.
- Collaborate on Git-based workflows, debugging, deployment, and production maintenance.

Projects

AI PDF Chatbot – End-to-End RAG Pipeline

LangChain, FAISS, FastAPI, Ollama, AWS | GitHub

- Architected a production-grade RAG pipeline: document ingestion → chunking → embedding generation (FAISS) → semantic retrieval → LLM synthesis via Ollama; achieved sub-2s end-to-end latency.
- Applied token-efficient prompting and context compression, reducing LLM API token usage by ~40%; engineered multi-turn memory and async processing, cutting latency by 35%.

AI Stock Sentiment Analyzer – LLM-Powered Intelligence Pipeline

FinBERT, ChromaDB, Groq API, Streamlit | GitHub

- Designed NLP pipeline: real-time news scraping → FinBERT classification (82% accuracy, 100+ articles/day) → ChromaDB vector storage → LLM market summaries via Groq API.
- Achieved 45% throughput gain via parallel fetching, embedding caching, and batched LLM calls.

Movie Recommendation System – Content-Based ML Engine

Scikit-Learn, FastAPI, TF-IDF, PostgreSQL, TMDb API | GitHub

- Built a content-based recommendation engine over 45,000+ movies using TF-IDF vectorisation and cosine similarity on plot, genre, and metadata; served via FastAPI with live TMDb poster data.

Skills

AI / GenAI	RAG Pipeline Design, LLM Integration, Prompt Engineering, Semantic Search, Vector Embeddings, Model Evaluation, AI Agents		
Deep Learning	TensorFlow, Keras, Neural Networks, CNNs, Backpropagation, Model Training & Evaluation		
ML / NLP	Scikit-Learn, FinBERT, HuggingFace Transformers, TF-IDF, Classification, NumPy, pandas, matplotlib		
AI Frameworks	LangChain, HuggingFace, Ollama, Groq API, OpenAI API, Anthropic Claude API		
Vector DB	FAISS, ChromaDB, Embedding Pipelines		
Web / Backend	FastAPI, Streamlit, React.js, Node.js, Express.js	Databases: PostgreSQL, MongoDB, Redis	
Cloud / Tools	AWS (EC2, S3), Docker, Git, GitHub, Jupyter Notebook	Languages: Python, java, SQL, JavaScript	

Education

RVS College of Engineering and Technology

B.Tech in Computer Science and Engineering | CGPA: 7.0/10

Jamshedpur

Nov 2022 - May 2026

Certifications

- Oracle Generative AI Professional Certification
- Oracle Cloud Infrastructure (OCI) Foundations Associate
- HuggingFace Fundamentals of Large Language Models
- Google Cloud Introduction to Generative AI
- TensorFlow Deep Learning Fundamentals
- IBM: How Large Language Models Work
- IBM: Natural Language Processing and Computer Vision
- JP Morgan Quantitative Research Virtual Experience (Forage)